

**AN INTELLIGENT SYSTEM FOR DETECTION,
TRANSLATION, AND HISTORICAL MAPPING OF
ANCIENT STONE INSCRIPTIONS FROM THE
ANURADHAPURA KINGDOM SRILANKA**

25-26J-144

Project Proposal Report
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August 2025

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Using image processing technology and artificial intelligence systems, this section focuses on improving the quality of raw stone office work and enhancing the accuracy of identifying areas exposed to coverage and damage. The ancient Brahmi inscriptions in the Anuradhapura Kingdom are significantly affected by soil erosion, degradation, and weathering, as well as the additional morphological features of some crucial elements that make field research especially challenging. This module uses images like superior proposals based on interpretive histogram matching, unprotected morphological features, and deep learning. Meanwhile, the data set described is used to train a system to recognize and categorize various damages such as artifacts, deletions, and missing parts in images. The selected transformations for the preservation of the documented occurrences may be parts of images that could damage aspects like facial features, character segmentation, recognition, and translations, as well as transformations that can incur costs. Full processing results in enhanced images with extensive damage that include facial features, thereby improving the quality of analyses involving AI and human specialists. This section enhances the capability of preserving cultural heritage and prevents data acquisition through better representation and transformation in preservation modeling. Automation allows for translation and regulation. This feature of the system is developed to fit the overall documentation system, thereby facilitating better digital acquisition and preservation of ancient Brahmi scripts throughout Sri Lanka.

Keywords:

image processing, artificial intelligence, stone inscription enhancement, Brahmi inscriptions, Anuradhapura Kingdom, damage detection, morphological analysis, histogram matching, deep learning, artifact classification, character segmentation, digital preservation, cultural heritage, automated annotation, restoration modeling, data acquisition, translation automation.

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1. INTRODUCTION

1.1 Background

Ancient Sri Lankan inscriptions have deep historic, cultural, and linguistic relevance. As well as precious information on the growth of cultures as early as before the time, in various regions of our island such as Anuradhapura kingdom they are. Engraved text or designs are carved or etched on the hard surface of stone. Inscription erodes throughout this, and often nature and human sensitivities such as interruptions, natural growth, and weather. This results in erosion of the material itself, which ultimately affects the legibility of the actual original copy, complicates interpretation of meaning, and makes it difficult to preserve.

As when one tries to study an inscription will be drawn into significant headstones due to blurred images, faded or deteriorated text, eroded areas, and indistinct photos. If the Traditional manual technique is used for inscription survey has many disadvantages such as sensitive to subjectivity, low scalability and totally depends on the archeologist. Recent advances in digital image processing provide solutions to these challenges. The image enhancement processes such as focusing on image optimization, reclamation, and automated cell location allow researchers to recover lost textual content and accurately recognize damaged areas.

Machine learning and modern computer vision solutions also offer abilities for quick non-destructive pre-processing, visualization, and damage cell analysis today. By strengthening the basics of these non-destructive secrets and image mapping, a significant contribution to precise archival investigation and preservation programs is made. However, there are several precise and distinct approaches for handling old stone inscription and various types of physical damage that need to be considered.

In this project expected to restore the brightness in ancient signs through advanced image processing and artificial intelligence. Provide the core element to be replicated and edited and create a good footing for future study in the study of history while also determining physical harm.

1.2 Literature Survey

A variety of studies have investigated the digital image processing of ancient texts to enhance their secrets and identify signs of deterioration.

For instance, the study “Restoration of Ancient Stone Inscriptions through Image Enhancement Techniques” conducted by J.Y. Lee, R. Srinivasan, and M. Karunaratne in 2022 shows the effectiveness of image sharpening techniques, such as unsharp masking and Laplacian filters, in enhancing the unrecognized characters of an inscription. This study demonstrates the value of preprocessing such as character recognition and translation which helps to further analysis. This study shows how sharpened ancient scripts lead to sharpening over 20 percent improvement in accuracy in OCR derived texts as against raw unprocessed texts.[\[1\]](#)

There has also been progress in the field of enhancement of inscriptions such as “Efficient Scalable Template-Matching Technique for Ancient Brahmi Character Recognition” by S. Chatterjee et al. in 2022, in which the sharpening of images, noise, and contrast provided the backbone of the advanced template matching systems designed for ancient scripts. Sharpened images made the feasible accurate extraction of historical scripts.[\[2\]](#)

The automation of identifying damage on ancient stones surfaces was studied in 2021 by T. Gupta, L. Rathnayake, and H.K. Pathirana, titled, “Detection and Mapping of Damage in Epigraphic Images Using Deep Learning.” This study trained convolutional neural networks (CNN) on annotated datasets with various types of damage such as cracks, abrasions, and biotic damage. Their model achieved a damage prediction and localization accuracy of 85%, which was better than manual techniques. This study demonstrates the important role of semantic segmentation models and in this study creating detailed damage masks that assist in the restoration and digital archiving of artifacts.[\[3\]](#)

Examining stone inscriptions, “Image Processing Approach for Ancient Brahmi Script Analysis” (Academia.edu, 2024) focused on the ancient Tamil inscriptions. And in this study, they are applying adaptive histogram equalization and multi-scale edge detection. The authors reported significant enhancement on the character visibility and recognition accuracy. This study showcased the potential of enhancement and damage detection techniques, and they focused on improving the readability and interpretability of inscriptions on stones in the context of heritage preservation.[\[4\]](#)

A different study was done in 2023 by Priya Raj and others called “Automated Preprocessing and Damage Annotation for Cultural Heritage Inscriptions.” This study focuses on the problem of having to deal with the image quality and the surrounding environment of the photograph’s inscriptions taken in the field. This study evaluated multiple different preprocessing pipelines combining noise reduction with adaptive histogram equalization and multi-scale edge detection to optimize what is both sharpened and analyzed for damage. The authors note that preprocessing not only helps to improve reading with the human eye but also helps in the performance of machine learning algorithms for segmentation and text recognition done later. The authors emphasize the importance of combining image sharpening and damage annotation long before the stage of performing digital epigraphy.[\[5\]](#)

1.3 Research Gap

Acquisition of Inscription Images of High Quality

The current approaches in obtaining images often capture images across different environments in uncontrolled lighting, focus, and resolution. This inconsistency hinders effective processing. This indicates that there is a need to create image capture protocols or preprocessing methods that can automatically improve raw inscription images irrespective of the capturing environment.

Image Sharpening for Legible Eroded and Faint Texts

Unsharp masking and Laplacian filters are common sharpening methods. They are frequently used to improve focus, but they fail to consider the distinctive surface wear and texture of stone inscriptions. Failure to consider this might result in the addition of artifacts or the erasure of crucial inscription details. More work is needed to devise algorithms that enhance images as prescribed to the specific features of ancient stone surfaces, increasing their readability and decreasing faking.

Automated and Robust Damage Spot Detection Systems

Manual damage detection takes forever. And it's super inconsistent because people see things differently. CNN models can spot cracks or erosion automatically, which sounds great, but here's the catch. They struggle with limited training data and how messy real-world damage types can get. The real challenge is that most datasets lack sufficient labeled samples. The challenge is, no one has figured out how to scale this for massive datasets of inscriptions that vary so much in appearance. You need an approach that manages diverse surface qualities and damage levels smoothly.

Table 1: System Comparisons

Features	[1]	[2]	[3]	[4]	[5]	System
Image enhancement for inscriptions	Yes	Yes	No	Yes	Yes	Yes
Automated damage detection	No	No	Yes	No	Yes	Yes
Integrated enhancement and damage detection	No	No	No	No	No	Yes
Adaptation to diverse inscription surfaces	No	No	Yes	No	Yes	Yes
Scalability to large heterogeneous datasets	No	No	No	No	No	Yes

1.4 Research Problem

Ancient stone inscriptions like those in Anuradhapura Brahmi scripts get beat up over time. Erosion cracks biological growth and all that stuff makes the carvings harder to see. The photos we have of these artifacts often look washed out grainy blurry too thanks to weather conditions and old-school camera methods. Basically, this combo makes it tough for experts to read what's written let alone analyze or preserve it properly.

Right now, the tools we use for cleaning up images aren't cut out for stone textures specifically. They either sharpen things wrong leave artifacts behind or miss damage spots entirely. Different methods get used separately which doesn't help much either. Manual checks take ages human error creeps in scaling up for large archives? Forget about it.

So here's the challenge to make a sharpening technique that works on worn stone surfaces without messing up the details further. Pair that with automated damage detection that can pinpoint cracks erosion spots whatever else is going on. Then tie both systems together into one workflow that handles tons of inscriptions at once. Get this right and suddenly those ancient texts become way clearer easier to study simpler to save digitally long-term. Big win for historians' archaeologists, anyone trying to decode old civilizations really.

2. OBJECTIVES

2.1 Main Objective

Establish and implement effective image sharpening and automated damage spots detection system to improve the legibility of ancient stone inscriptions and accurately identify physical damage regions. This will allow for better epigraphic research and digital preservation.

2.2 Specific Objectives

1. Enhance inscription image clarity

Apply advanced image sharpening processes like adaptive filtering, contrast adjustment, and edge enhancement to improve the legibility of weak or deteriorated text in raw inscription images.

2. Automate damage spots detection

Develop and train machine learning models like convolutional neural networks (CNNs) to detect physical damage like cracks, erosion, and biological marks on stone inscriptions & segment them.

3. Produce reliable damage masks

Provide accurate annotations of damaged spots for all images and provide critical inputs to downstream restoration and script recognition tasks.

4. Optimize preprocessing pipeline

Create a seamless, scalable pipeline that can handle various inscription images obtained under various acquisition conditions

5. Improve usability and visualization

provide visualization tools to display damaged areas and improved photos. to conservators and researchers engage with the findings.

3. METHODOLOGY

This component enhances ancient inscription images to sharpen clarity using noise reduction and contrast enhancement. It detects damaged or eroded areas through edge detection and deep learning-based segmentation. Producing damage masks that assist in preserving and restoring the inscriptions.

1. Image Acquisition

Capture high-resolution images or 3D scans of ancient Brahmi inscriptions using high quality cameras or scanners under controlled lighting to reduce shadows and reflections.

2. Preprocessing and Sharpening

- **Noise Reduction**

Apply filters such as Gaussian blur or median filtering to reduce noise while preserving edges.

- **Image Sharpening**

Use techniques like Unsharp Masking, Laplacian filtering, or High Pass filtering to enhance edge details and make inscriptions clearer.

- **Contrast Enhancement**

Utilize Histogram Equalization or Adaptive Histogram Equalization (CLAHE) to improve visibility of faded inscriptions.

- **Binarization**

Convert images to binary form by applying thresholding techniques (Otsu's method) to separate inscription text from background.

3. Geometric Corrections

Fix geometric errors and perspective issues with image transformations (rotation, scaling, perspective warp) of normalized and aligned images.

4. Damage Detection

Extract texture and edge features using methods like Canny Edge Detection, Sobel filters, or Local Binary Patterns.

Use morphological operations (dilation, erosion) to highlight damaged regions.

Employ supervised machine learning or deep learning algorithms for damage segmentation:

- **Classical ML**

Random Forest, SVM based on extracted features.

- **Deep Learning**

Train Convolutional Neural Networks (CNN) or U-Net segmentation models on annotated datasets of damaged vs. intact inscription images.

Generate damage masks highlighting cracked, eroded, or missing parts in the inscriptions.

5. Post-Processing and Validation

Refine masks with conditional random fields (CRF) or morphological cleaning.

Visualize and validate detected damage areas with domain experts and iteratively improve models with user feedback.

Table 2: Tools & Technologies

Task	Tools & Technologies	Description
Image processing and sharpening	OpenCV, Scikit-Image, PIV 	Standard Python libraries for filtering, sharpening, and contrast enhancement
Geometric correction	OpenCV	Image warping and alignment functions
Feature extraction	OpenCV, scikit-image	Includes edge detectors (Canny, Sobel), texture analysis (LBP)
ML model training	TensorFlow, PyTorch, scikit-learn 	For training CNNs/U-Net or classical ML models
Annotation and visualization	Labelbox, CVAT, Custom annotation tools 	For labeling damaged regions in images
Validation and collaboration	Jupyter notebooks, Dash, Flask 	Visualization frameworks and interactive tools

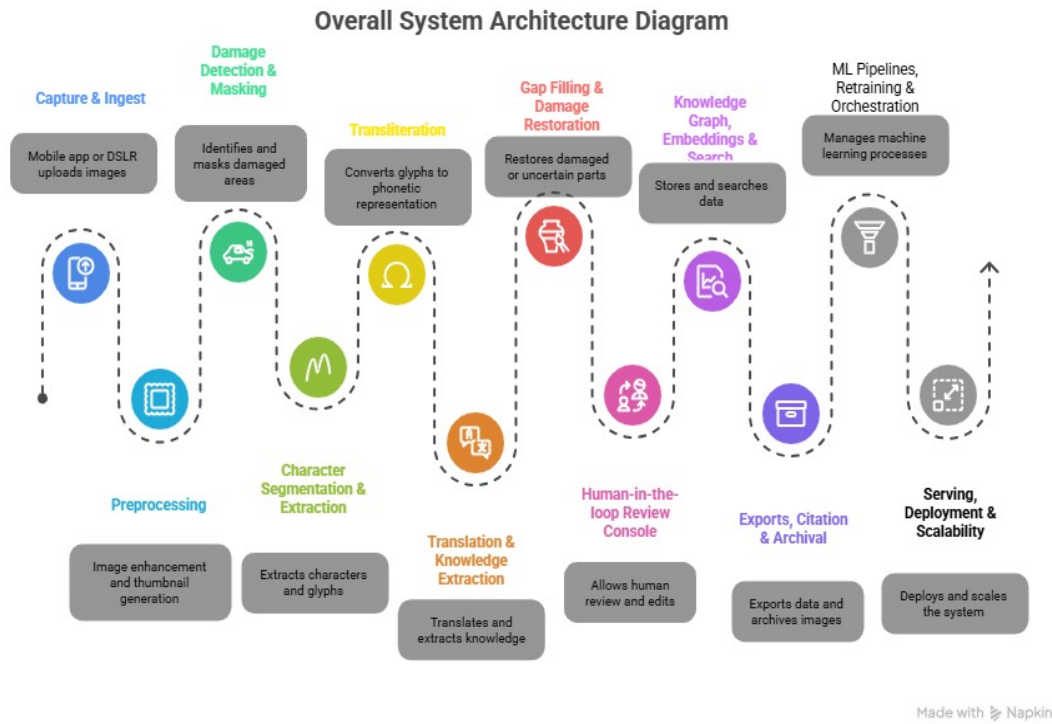


Figure 1: Overall System Diagram

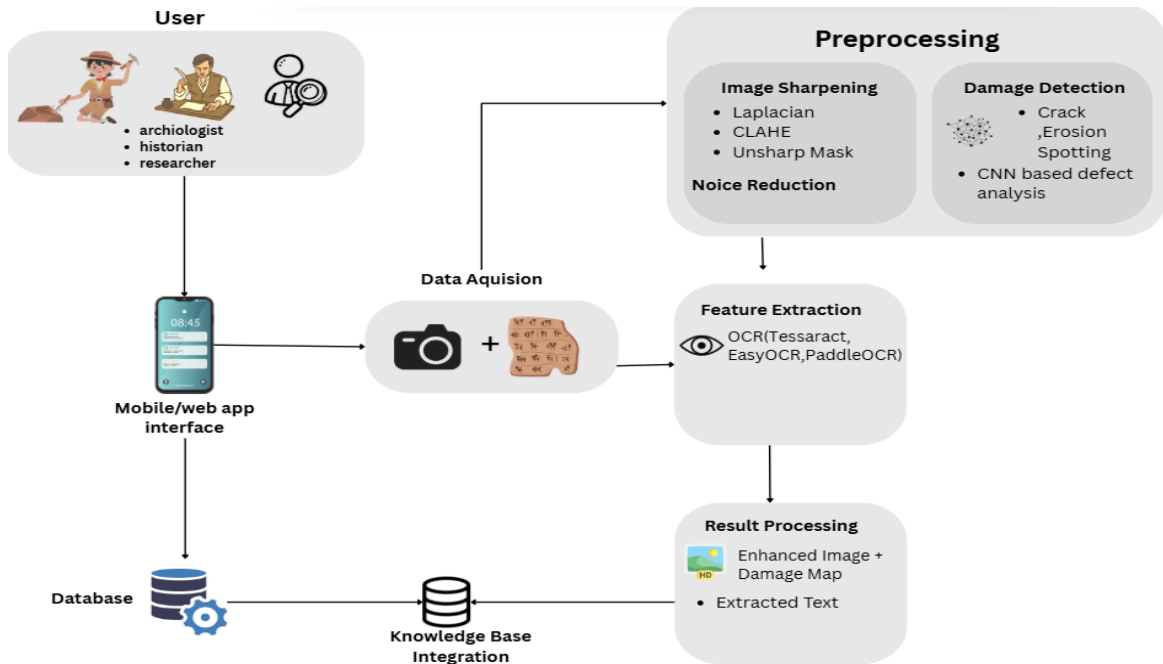


Figure 2: System Diagram for Image Sharpening & Damage Detection

4. PROJECT REQUIREMENTS

Data Collection

The required information will be gathered through archaeological images databank, real-life surveys, and digital archives of inscriptions related to heritage. Annotations to the damage spots will be done by domain experts or crowdsourcing platforms, thus providing quality ground truth to model training and validation.

Data Source 1: The digital archives committed by the Archaeological Department

Data Source 2: Inscription image datasets publicly available over with damage annotations

4.1 Functional Requirements

Image Input

The system will have the capability of accepting high resolution images of stone inscriptions in different formats (e.g., JPG, TIFF, PNG).

Image Preprocessing

The system will include noise removal, normalization of brightness/contrast, and orientation realignment.

Image Sharpening

The algorithm will run adaptive sharpening to make text easier to read but retaining the look of the surface.

Damage Detection

The system will automatically detect damaged areas (cracks, erosion, biological growth) and produce damage masks.

Visualization

The system will allow providing visualization overlays with enhanced images with annotations of damage.

Export Capability

The system will be able to export the processed images, and damage masks as a form of analysis to other systems.

Batch Processing

The system will support batch processing several images and thus make the system scalable.

4.2 Non-Functional Requirements

Availability

The system would be accessible within normal working hours at a minimal level of downtimes.

Performance

The system must have high performance on how it processes images, the average time taken on each image must be processor friendly.

Reliability

The results must be same and correct with error checking on unsupported formats or corrupt data.

Scalability

It must be able to support increased traffic of images with no deterioration of response time and accuracy.

Usability

The applications ought to be user-friendly, and have interfaces that are understandable, yet not based on technological knowledge.

Accessibility

The system must be accessible according to accessibility standards; it must be usable by people with disabilities.

4.3 The Expected Test Cases

- Processing pictures of various resolutions and lightening.
- Confirmation of sharpening effect on dark and strong eroded text.
- The use of expert-annotated masks and a correctness test of damage detection.
- The mixed image quality Batch processing capability.
- Functionality and compatibility of user interface and correctness of visualization.

5. BUDGET & BUDGET JUSTIFICATION

Table 3: Budget

Requirement	Cost (Rs.)
Fieldwork/Travel for Data Collection	Rs. 25,000.00
Purchasing virtual storage	Rs. 4,000.00
Software Licenses (OCR/Image Tools)	Rs. 6,000.00
Miscellaneous (app store fees etc.)	Rs. 5,000.00
Hosting	Rs. 5000.00 per month
Total Cost	Rs. 45,000.00 + monthly hosting fee

Justifications

- **Fieldwork/Travel for Data Capturing (Rs. 25,000.00)**

This is for visiting sites and capturing real images of inscriptions. It covers transportation, travel, and modest expenses during data capturing.

- **Purchasing Virtual Storage (Rs. 4,000.00)**

Since images take up a lot of space, more space is needed to safely store all the data and make it readily available to members of the team.

- **Software Licenses (OCR/Image Tools) (Rs. 6,000.00)**

Special software will be required for image and text processing. The license cost will allow us to make use of advanced features for accurate results

- **Miscellaneous (Rs. 5,000.00)**

This is kept for small or unexpected costs such as app store fees or other minor necessities that might crop up.

- **Hosting (Rs. 5,000.00/month)**

The system should be hosted online so that it is readily accessible. The monthly subscription will maintain the system operational and always in service.

REFERENCES

- [1] J. Y. Lee, R. Srinivasan, and M. Karunaratne, "Restoration of Ancient Stone Inscriptions through Image Enhancement Techniques," *Int. J. Adv. Res. Sci., Commun. Technol.*, 2022, doi: 10.48175/IJARSCT-3090. [Online]. Available: <https://ijarsct.co.in/Paper3090.pdf>
- [2] S. Chatterjee, P. K. Roy, and A. Majumder, "Efficient scalable template-matching technique for ancient Brahmi character recognition," *ScienceDirect*, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1546221822002028>
- [3] T. Gupta, L. Rathnayake, and H. K. Pathirana, "Detection and mapping of damage in epigraphic images using deep learning," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLII-2/W15, pp. 639–644, 2021. [Online]. Available: <https://isprs-archives.copernicus.org/articles/XLII-2-W15/639/2019/>
- [4] Anonymous Authors, "Image processing approach for ancient Brahmi script analysis," *Academia.edu*, 2024. [Online]. Available: https://www.academia.edu/117632272/Image_Processing_Approach_for_Ancient_Brahmi_Script_Analysis
- [5] P. Raj *et al.*, "Automated preprocessing and damage annotation for cultural heritage inscriptions," *J. Cult. Herit. Preserv.*, 2023.